DAB Reference Paper & Proposed 3 kW / 30 kHz Work Plan

1. Reference IEEE Paper

Paper: A Low Complexity Algorithm for Efficiency Optimization of Dual Active Bridge Converters
Authors: Simone Pistollato, Nicola Zanatta, Tommaso Caldognetto, Paolo Mattavelli
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2. Reference Paper Methodology

The reference paper proposes an efficiency optimization method for a Dual Active Bridge (DAB) converter using Triple Phase Shift (TPS) modulation and Maximum Efficiency Point Identification (MEPI). TPS uses the modulation variables D1, D2 and phase shift ϕ. The method aims to reduce RMS current, improve the ZVS operating region and optimize converter efficiency. The paper gives particular attention to light-load efficiency and extends the optimization procedure to the complete output-power range.

3. Reference Paper vs. Proposed Work

Parameter	Reference Paper	Proposed Work
Converter	DAB	DAB
Experimental power	1.5 kW prototype	3 kW
Switching frequency	Paper prototype conditions	30 kHz
Modulation	TPS	TPS
Optimization	MEPI	MEPI
Purpose	Efficiency optimization	Adaptation, validation and further research

Key distinction: The reference paper experimentally validates its methodology using a 1.5 kW DAB prototype. The proposed work will implement and adapt the same TPS + MEPI methodology for a 3 kW, 30 kHz DAB converter.

4. Proposed Implementation Sequence

1. Develop the 3 kW, 30 kHz DAB converter model in MATLAB/Simulink.

2. Implement the TPS modulation proposed in the reference paper using D1, D2 and ϕ.

3. Analyze ZVS operation, transformer current, RMS current and transferred power.

4. Implement the MEPI-based efficiency optimization methodology proposed in the reference paper.

5. Evaluate efficiency and compare the observed operating trends with the reference paper.

6. Validate the complete TPS + MEPI methodology for the 3 kW, 30 kHz operating conditions.

7. After baseline validation, analyze limitations and identify the research gap.

8. Develop and validate a suitable innovative extension based on the identified gap.

5. Research Direction

The first objective is to establish a validated baseline by adapting the reference paper's TPS + MEPI methodology to the proposed 3 kW, 30 kHz DAB converter. Innovation will be defined after identifying a specific limitation or performance gap during this validation, rather than assuming a novelty before the baseline is established.

Work strategy: Reference paper (1.5 kW) → TPS + MEPI implementation → 3 kW, 30 kHz validation → limitation analysis → research gap → innovative extension.